

Prithvi Agro AI: An Evidence-Driven Multimodal Agricultural Intelligence Platform for Personalized Crop Advisory, Disease and Pest Detection, and Retrieval-Augmented Agricultural Assistance

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Abstract—

Agricultural decision-making depends on a wide mix of factors — soil composition, weather, crop suitability, pest and disease pressure, market conditions, and locally accumulated agronomic knowledge. Most existing agricultural software tools handle these factors in isolation: a weather app shows weather, a market app shows prices, and a chatbot answers questions with no real evidence behind its claims. Farmers are left to stitch these fragments together themselves. This paper presents Prithvi Agro AI, an agricultural intelligence platform designed around a different premise — that farm advisory should be personalized to a specific farm's context, grounded in retrievable evidence, and explicit about what it does not know. The system combines structured farm records held in PostgreSQL, relationship-based agricultural knowledge in a Neo4j graph, and semantic document retrieval through a vector database, feeding all three into a retrieval-augmented generation layer before a language model produces any response. Separate machine-learning pipelines handle crop recommendation from soil and climate parameters, and computer-vision models classify pest and disease images from field photos. Weather, historical climate, and Sentinel-2 satellite observations are integrated as live, provenance-tagged data sources rather than static content. A safety and evidence layer sits in front of every generated response, attaching confidence levels, cited sources, and freshness indicators, and allowing the system to abstain rather than answer when the underlying evidence is too thin to support a claim — a deliberate design choice given that agricultural misinformation, particularly around pesticide use, carries real financial and safety consequences. This paper describes the complete system architecture, the reasoning behind combining three distinct data-retrieval paradigms rather than relying on one, the evaluation methodology intended for the machine-learning, retrieval, and language-generation components separately, and the limitations of the design as currently specified. The system is described at the architectural and pre-deployment stage; no field results are reported here.

Keywords—*agricultural intelligence, retrieval-augmented generation, knowledge graph, precision agriculture, computer vision, crop recommendation, plant disease classification, satellite remote sensing, explainable AI, multimodal systems*

I. INTRODUCTION

A farmer deciding what to plant this season is really answering several different questions at once. What does the soil support? What has the weather been doing, and what is it likely to do over the next few weeks? Has this crop had pest or disease problems in this region before? What will it sell for at harvest, and which market is worth the trip? None of these questions has the same shape — one is a numerical prediction problem, another is a visual classification problem, another calls for retrieving and interpreting agronomic literature, and another depends on live market data. Answering all of them well requires several different kinds of system working together, not one generalist tool.

Most agricultural software available today does not work this way. Weather apps show weather. Market-price apps show prices. A growing number of chatbot-style assistants attempt to answer open-ended farming questions, but the answers frequently arrive without any indication of where the information came from, how confident the system actually is, or whether the advice applies to the specific field being discussed. For a domain where a wrong recommendation can mean a lost harvest or an unsafe pesticide application, that lack of traceability is a real problem, not just an inconvenience.

Large language models have made conversational agricultural assistants more feasible to build, but a model answering purely from what it learned during training is prone

to the same failure mode seen across other high-stakes domains — confident, fluent, and occasionally wrong in ways that are hard to catch. Retrieval-Augmented Generation is the standard mitigation, grounding a model's output in retrieved documents rather than parametric memory alone. Even so, a single retrieval pipeline pointed at a single kind of data store cannot really answer all the question types listed above. A question about a farm's own recorded soil pH is a database lookup. A question about which pests have historically affected crops in a particular region is a relationship query, better suited to a graph structure than a flat table or a document index. A question asking what agronomic literature recommends for a disease is a semantic search problem. Treating all three as if they were the same kind of retrieval tends to produce mediocre results across the board.

This paper describes Prithvi Agro AI, an agricultural intelligence platform built around this observation. Rather than routing every question through one retrieval mechanism, the system maintains three distinct and complementary data layers — a PostgreSQL store for structured, authoritative farm records; a Neo4j knowledge graph for relationship-based agricultural reasoning; and a vector database for semantic retrieval over agricultural documents and advisories. A retrieval orchestration layer decides, per query, which of these sources are relevant, and a context-fusion step combines what comes back before any language model reasoning happens.

Machine-learning models handle crop recommendation from soil and climate inputs; computer-vision models handle pest and disease classification from field images; and live environmental data — weather, historical climate, and Sentinel-2 satellite imagery — feed into the same evidence pipeline rather than sitting off to the side as separate app features.

A safety and evidence layer sits in front of the language-model output on every request. Instead of letting the model state a conclusion unconditionally, the system attaches a confidence level, a list of the specific evidence used, and freshness indicators showing how current the underlying data is. When the available evidence does not clearly support an answer, the system is designed to say so rather than fill the gap with a plausible-sounding guess. This matters most acutely around pesticide and chemical recommendations, where an incorrect suggestion driven by a low-confidence pest classification could cause real crop damage or environmental harm — a case discussed further in Section VII.

The contributions of this paper are as follows:

- 1) A hybrid retrieval architecture combining structured SQL data, graph-based relationship reasoning, and semantic vector search under one orchestration layer, routed per query rather than applied uniformly.
- 2) A farm-centric context model that keeps every environmental, agronomic, and advisory response scoped to a specific farm's actual conditions rather than generic regional averages.
- 3) An evidence and provenance framework that attaches source, confidence, and freshness metadata to every generated response, including an explicit abstention pathway for low-confidence cases.
- 4) Integration of machine-learning crop recommendation, computer-vision pest and disease classification, and satellite-based vegetation monitoring into one coherent evidence pipeline rather than as separate disconnected tools.
- 5) A proposed evaluation methodology spanning the machine-learning, retrieval, and generation components separately, including hallucination-rate and evidence-coverage metrics specific to the agricultural advisory setting.

The rest of the paper proceeds as follows. Section II covers related work in agricultural recommendation systems, plant disease detection, agricultural chatbots, and retrieval-augmented generation. Section III lays out the specific research gap this work targets. Section IV presents the system architecture in full, including the hybrid retrieval design and the individual ML and vision pipelines. Section V covers the safety and evidence architecture. Section VI describes the data sources and implementation. Section VII discusses limitations, Section VIII covers future work, and Section IX concludes.

II. RELATED WORK

Relevant prior work falls into five loosely connected areas: agricultural recommendation systems, plant disease and pest detection through computer vision, agricultural chatbots and conversational assistants, retrieval-augmented generation, and

knowledge-graph-based reasoning. Progress in each area individually has been substantial. Work that combines them into one coherent farm-facing system is comparatively rare.

A. Agricultural Recommendation Systems

Crop recommendation from soil and climate parameters is a well-studied machine-learning problem. Systems using Random Forest, XGBoost, and similar ensemble methods on inputs like nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall have reported strong classification accuracy across several published datasets [1], [2]. These systems are generally standalone — a farmer enters soil values and receives a ranked list of suitable crops, without that recommendation being tied into weather forecasts, satellite-observed field conditions, or the farm's own planting history. Doshi et al. [3] and Pudumalar et al. [4] both note this limitation directly, pointing out that recommendation quality would improve if the model had access to real-time environmental context rather than static input values entered by hand.

B. Plant Disease and Pest Detection

Convolutional neural networks trained on datasets like PlantVillage [5] have become the standard approach for plant disease classification from leaf images, and reported accuracy figures on held-out PlantVillage test sets are often above 95%. That number is somewhat misleading in practice, though — Mohanty et al. [5] themselves note a sharp accuracy drop when the same models are tested on field-captured images rather than the controlled, lab-photographed images PlantVillage largely consists of. Similarly, the IP102 dataset [6] for pest classification is explicitly designed around this problem, including a long-tailed class distribution and field-realistic image conditions specifically because lab-only training data does not generalize well. Ferentinos [7] reports comparable findings across several CNN architectures: the gap between benchmark accuracy and field accuracy is a persistent, acknowledged issue in this literature, not a solved one.

C. Agricultural Chatbots and Conversational Systems

Rule-based and retrieval-based agricultural chatbots predate the current generation of LLM-based assistants by some margin, typically built around a fixed FAQ or decision-tree structure covering common farming questions [8]. More recent LLM-based agricultural assistants improve on the breadth of questions they can handle but tend to inherit the general-purpose hallucination problem of large language models — Kasneci et al.'s broader survey on LLMs in specialized domains [9] flags agriculture, alongside medicine and law, as a domain where unsupported generated claims carry outsized real-world consequences. Very few of the agricultural chatbot systems in the literature report any structured evidence or provenance mechanism; most simply generate a conversational answer directly.

D. Retrieval-Augmented Generation and Knowledge Graphs

Retrieval-Augmented Generation [10] grounds model output in retrieved documents rather than parametric memory alone and is now the standard technique for reducing hallucination in

knowledge-intensive generation tasks. Most RAG implementations, agricultural or otherwise, use a single vector index as the retrieval source. GraphRAG approaches [11], [12] extend this by adding graph-structured retrieval alongside or instead of vector search, on the reasoning that some questions are fundamentally about relationships between entities — which this crop had which pest, in which region, under which conditions — and that a graph traversal answers this kind of question more directly than a semantic similarity search over unstructured text. Edge et al. [11] specifically show that graph-based retrieval outperforms flat vector RAG on multi-hop reasoning questions, which is relevant here since a farmer's question ('why are my leaves yellow after this much rain, given what happened to my neighbor's field last season') is often exactly this kind of multi-hop question. Hybrid architectures combining SQL, graph, and vector retrieval together, as opposed to any two of the three, are not well represented in the published literature — this is the specific combination Section IV proposes.

E. Satellite and Remote-Sensing Agriculture

NDVI and other vegetation indices derived from Sentinel-2 and similar satellite platforms are widely used for crop-health and vegetation monitoring at scale [13], and open access to Sentinel-2 data through platforms like Microsoft's Planetary Computer has made satellite-based monitoring far more accessible to smaller projects than it was a decade ago. Most published work in this area treats satellite monitoring as a standalone analytics product — a dashboard showing NDVI trends over time — rather than as one evidence source feeding into a broader, personalized advisory system alongside weather, soil, and pest data.

Table I summarizes where these five areas of prior work sit relative to what this paper proposes.

TABLE I.

Comparative Summary of Related Work

Area	Strength	Limitation Relative to This
Crop Recommendation ML [1]-[4]	Strong classification accuracy	No live environmental integrati
Disease/Pest CV [5]-[7]	High benchmark accuracy	Weak field-condition generaliza
Agricultural Chatbots [8],[9]	Broad question coverage	No evidence or provenance laye
RAG / GraphRAG [10]-[12]	Reduces hallucination, multi-hop reasoning	Not applied to farm-specific hyl retrieval
Satellite Monitoring [13]	Scalable vegetation tracking	Standalone, not fused with advi system
Proposed Platform	Hybrid SQL+Graph+Vector, evidence-gated, multimodal	Not yet field-evaluated

III. RESEARCH GAP AND MOTIVATION

The related work in Section II shows real progress in each individual area, but four gaps show up once the pieces are considered together.

First, recommendation and classification systems are typically built and evaluated in isolation from the live environmental context that would make them more useful. A crop-recommendation model trained on soil nutrient values

has no inherent connection to that day's weather or that field's satellite-observed vegetation state, even though a real recommendation should probably account for all three.

Second, agricultural chatbots built on large language models largely lack any structured evidence mechanism. A farmer asking why their crop looks unhealthy gets a fluent answer, but rarely gets a citation, a confidence level, or an indication of how current the underlying data is — three things that matter a great deal when the advice concerns a real field and a real season.

Third, hybrid retrieval combining structured (SQL), relationship-based (graph), and semantic (vector) data sources together is underexplored specifically in agriculture, despite GraphRAG work elsewhere showing real benefit for multi-hop reasoning questions. Agricultural questions are frequently multi-hop by nature — a question about yellowing leaves plausibly depends on crop identity, recent weather, regional pest history, and agronomic literature all at once — which makes this combination a reasonably natural fit that the literature has not applied here yet.

Fourth, computer-vision pest and disease classifiers are commonly deployed without an explicit accounting for the accuracy gap between lab-condition benchmark data and field-captured images, and without a clear policy for what the system should do when classification confidence is low — particularly where that classification could otherwise lead to a chemical treatment recommendation.

These four gaps motivate the design choices in Section IV: a hybrid three-source retrieval architecture routed per query rather than a single retrieval mechanism; a farm-centric context model that ties every response to a specific farm's actual live conditions; an evidence and confidence layer attached to every generated response, including explicit abstention; and a computer-vision pipeline that treats low-confidence classifications as a reason to withhold chemical recommendations rather than a caveat attached after the fact.

IV. PROPOSED SYSTEM ARCHITECTURE

A. Architecture Overview

The system is organized around a central principle: no response should reach the farmer without a traceable connection to the specific data that produced it. The architecture has six layers. A web interface layer (dashboard and a conversational assistant, described further in Section IV-H) accepts farmer queries. An API gateway routes requests. Three parallel retrieval layers — structured (PostgreSQL), relational (Neo4j), and semantic (vector database) — each hold a different kind of agricultural knowledge. A context and RAG orchestration layer decides which of the three sources a given query actually needs and fuses what comes back. Machine-learning, computer-vision, and external environmental APIs feed additional evidence into the same fusion step. A safety and evidence layer sits between the fused context and the language model, and gates what the model is allowed to state unconditionally versus what requires a citation or a confidence qualifier.

Figure 1 shows this as a data-flow diagram: a query enters through the web interface, is routed by the API gateway, triggers parallel retrieval across the three data layers plus any relevant ML/vision/external-API calls, gets fused into one evidence context, passes through the safety and evidence layer, and only then reaches the language model for a grounded, cited response.

Fig. 1. High-level system architecture.

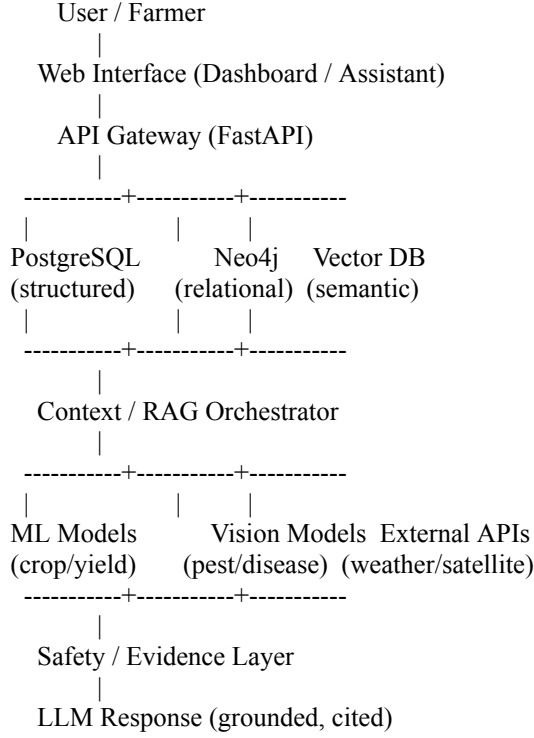


Fig. 1. Data flow from farmer query to evidence-grounded response.

B. Why Three Retrieval Systems

A recurring design decision in this architecture is the refusal to make one database solve every retrieval problem. Each of the three retrieval layers is suited to a different question shape, and forcing all questions through one of them produces a worse answer than routing each to the layer built for it.

PostgreSQL holds the system's authoritative structured data — user accounts, farm records, soil measurements, crop history, and inference logs. Questions with a direct factual answer sitting in a specific field of a specific record ("what is my farm's recorded soil pH") are SQL lookups, and treating them as anything more elaborate would only add latency and retrieval noise.

Neo4j holds agricultural knowledge as a graph of entities and relationships — farmer OWNS farm, farm HAS_SOIL soil-type, farm GROWS crop, crop EXPERIENCED pest, pest RELATED_TO region. Questions that are fundamentally about relationships ("which pests have affected crops in this region before") are graph traversals, and a graph structure answers them far more directly than either a table scan or a semantic search over unstructured text would.

The vector database holds agricultural documents, advisories, and research material as dense embeddings for semantic retrieval. Questions asking what established

agronomic guidance says about a topic ("what does ICAR recommend for managing this disease") are a document-retrieval problem best solved with semantic similarity search, not a database query or a graph traversal.

Table II summarizes this routing logic with representative query examples.

TABLE II.

Query Type to Retrieval Layer Mapping

Query Type	Layer	Example
Structured fact	PostgreSQL	Farm's recorded soil pH
Relationship	Neo4j	Pests seen in this region
Knowledge / advisory	Vector DB	ICAR guidance on a disease
Multi-hop / complex	All three + fusion	Why are leaves yellowing after heavy rain

C. Retrieval Orchestration and Context Fusion

A query first passes through intent detection and classification, which produces a rough estimate of which retrieval layers are relevant. Simple factual questions typically route to one layer. Complex or personalized questions — the yellowing-leaves example used earlier is representative — trigger parallel retrieval across all three, since the answer plausibly depends on the farm's own recorded data, regional pest relationships, and general agronomic knowledge simultaneously.

Algorithm 1 sets out this routing and fusion logic.

Algorithm 1: Query Routing and Context Fusion

Input: user query q , active farm f
Output: fused evidence context C

```

1: intent ← ClassifyIntent(q)
2: L ← SelectLayers(intent) // subset of {SQL, Graph, Vector}
3: results ← {}
4: for layer l in L in parallel do
5:   results[l] ← l.Retrieve(q, f)
6: end for
7: if RequiresLiveData(intent) then
8:   results[env] ← FetchEnvironmentalAPIs(f)
9: end if
10: if RequiresInference(intent) then
11:   results[ml] ← RunMLorVisionModel(q)
12: end if
13: C ← FuseContext(results)
14: C ← FilterByRelevance(C)
15: if C.evidence_coverage <  $\theta_{min}$  then
16:   C ← RequestAdditionalRetrieval(C, L)
17: end if
18: return C
  
```

Line 15 is the coverage check central to the safety design discussed further in Section V: if the fused context does not clear a minimum evidence-coverage threshold θ_{min} , the orchestrator requests another retrieval pass rather than letting the language model reason over a thin evidence base. This mirrors the logic used in the coverage-gated multi-agent frameworks described in recent agentic-RAG literature,

adapted here to a three-source retrieval setting rather than a multi-agent one.

D. Crop Recommendation Pipeline

The crop recommendation module takes standard soil and climate inputs — nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall — and produces a ranked list of suitable crops with associated probabilities. The candidate model families under consideration are Random Forest, XGBoost, and LightGBM, consistent with what the recommendation literature in Section II-A reports as strong performers on this kind of tabular data. Feature-importance is intended to be surfaced through SHAP values, so that a recommendation is not just a ranked list but comes with some indication of which input most influenced the result — useful both for farmer trust and for catching cases where a single unusual input value is driving an otherwise implausible recommendation.

Critically, this module does not operate on user-entered values alone. Because the farm context (Section IV-F) already holds recorded soil data and the environmental integration (Section IV-G) provides live weather and historical climate figures, the recommendation pipeline can draw on live, farm-specific context rather than requiring the farmer to manually re-enter values the system already has. This directly addresses the isolation problem in standalone recommendation systems noted in Section III.

E. Computer Vision Pipeline

Pest and disease classification both follow the same general pipeline: image validation and preprocessing, inference through a trained vision model, and a confidence score alongside the top prediction and plausible alternative classes. Given the field-versus-lab accuracy gap discussed in Section II-B, the pipeline is explicitly designed to treat classification confidence as a gating variable rather than a footnote. A low-confidence result does not proceed to a chemical or treatment recommendation; instead the system either requests a clearer image or suggests consulting an agronomist, consistent with the safety principle detailed in Section V-C.

The response format for both pest and disease classification includes the predicted class, a confidence category (high, moderate, or low, rather than a raw and potentially misleading numeric score), a short list of alternative plausible classes, and an explicit statement that image-based classification is an AI-assisted estimate rather than a confirmed field diagnosis.

F. Farm-Centric Context Model

Every module in the system is scoped to an active farm rather than operating generically. A farmer with multiple registered farms selects one as active, and that selection propagates to weather, climate, soil, crop planning, satellite monitoring, pest and disease history, market data, and advisory retrieval simultaneously. Switching the active farm invalidates and refetches all farm-dependent state; the design explicitly avoids the failure case where, for instance, one farm's weather data gets paired with another farm's satellite data in the same response.

This context model is what makes the retrieval orchestration in Section IV-C genuinely personalized rather than generically agricultural. A question routed to the vector database for agronomic advisory content still gets filtered and re-ranked against the active farm's recorded crop, soil type, and region, rather than returning generic top-matching documents regardless of farm context.

G. Environmental and Satellite Intelligence

Weather data is sourced from Open-Meteo and includes current conditions, hourly and daily forecasts, precipitation, wind, and humidity. Historical climate data — distinct from weather, and used specifically for identifying long-term patterns rather than immediate forecasting — is sourced from NASA POWER. Both are integrated as live API calls tied to the farm's registered coordinates rather than static or manually updated content.

Satellite monitoring uses Sentinel-2 imagery accessed through the Planetary Computer STAC API. The pipeline searches for available scenes over the farm's coordinates and date range, filters by cloud cover, and computes NDVI and related vegetation indices from qualifying scenes, enabling a form of crop-health monitoring that does not depend entirely on farmer-reported field observations. Every environmental and satellite response carries a provenance object — source, retrieval timestamp, and a staleness flag — described further in Section V-B, so that the interface can distinguish live data from cached or aging data rather than presenting both identically.

H. Conversational Interface

A conversational assistant, referred to internally as Saheta, sits alongside the dashboard as an alternative interaction mode intended primarily for farmers who find a full dashboard interface harder to navigate than natural conversation. Saheta is designed to route through the same backend services as the dashboard — the same API gateway, the same authentication, the same active-farm context, the same RAG pipeline — rather than operating as a separate, disconnected system. A voice or text command such as "show today's weather" or "scan this crop" is intended to route to the same weather endpoint or pest-detection workflow the dashboard uses, keeping behavior consistent across both interaction modes. Multilingual support (English, Hindi, Marathi, and Hinglish) is planned at this interaction layer, since language choice should not require re-establishing farm context, session state, or conversation history.

V. SAFETY AND EVIDENCE ARCHITECTURE

A. Evidence Model

Every substantive claim the system generates is meant to be traceable to a specific piece of evidence rather than presented as an unattributed statement. An evidence item carries a kind (document, vision inference, or agronomic rule), a source, a reference, and a short supporting detail. This is a deliberate departure from typical chatbot output, where the user sees only the model's final answer with no way to check what it was based on. The practical effect is that a farmer, or an

agronomist reviewing the system's output, can ask not just what the system said but why it said it.

B. Provenance and Freshness

Environmental and satellite data specifically carry a provenance object recording the data source, the retrieval timestamp, the originating URL where applicable, and a boolean staleness flag. If `is_stale` evaluates true, the interface is required to surface this rather than silently presenting aging data as current. This distinction matters more in agriculture than it might elsewhere — a three-day-old satellite scene or a six-hour-old weather forecast can meaningfully misrepresent current field conditions, and a farmer acting on stale information without knowing it is stale is a worse outcome than the system simply saying the most recent available data is somewhat old.

C. Confidence and Abstention

The system uses categorical confidence levels — high, moderate, low — rather than raw numeric scores, on the reasoning that a bare probability figure is often harder for a non-specialist to interpret correctly than a plain-language category, and can convey false precision. Confidence values are never fabricated by the interface layer; if the backend does not supply a confidence figure for a given response, none is displayed.

When evidence is insufficient, confidence is low, or required data is unavailable, the system is designed to abstain rather than generate a plausible-sounding answer regardless. Practically, this means a response along the lines of "this could not be verified from the available agricultural sources," rather than a confident answer built on thin or absent evidence. This abstention pathway is treated as a first-class output state in the design, on the same footing as a normal answer, rather than as an error condition to be minimized or hidden.

D. Safety Around Chemical and Pesticide Recommendations

The most safety-sensitive path in the system is the one connecting a pest or disease classification to a chemical treatment recommendation, and this path is treated with the most caution. A low-confidence image classification is explicitly barred from proceeding to a specific chemical recommendation. Instead, the system either requests a clearer image or points the farmer toward an agronomist. The justification is straightforward: an incorrect pesticide recommendation driven by a misclassified image can cause direct crop damage, financial loss, and environmental harm, and the cost of under-recommending (asking for a better image, or a human check) is far lower than the cost of over-recommending on weak evidence.

E. Safety Pipeline

Figure 2 summarizes the full safety check sequence applied before any model output reaches the farmer.

Fig. 2. Safety and evidence gating sequence.

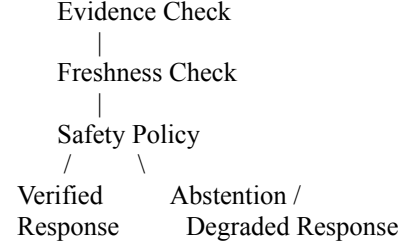
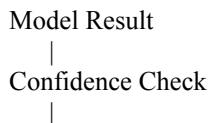


Fig. 2. Every generated response passes through confidence, evidence, and freshness checks before a safety policy decides whether to return a full answer or an abstention.

F. Data Isolation and Access Control

A separate but related safety concern is data isolation between users. The system enforces that a farmer can never receive another farmer's farm records, inference history, or personalized data, and this is intended to be enforced at the backend data-access layer rather than through interface-level hiding alone, since a frontend restriction is not a substitute for an actual authorization check. Role-based access — farmer, agronomist, admin — follows the same principle: route guards exist in the interface for usability, but backend authorization remains the actual source of truth for what a given request is allowed to do.

VI. DATA SOURCES AND IMPLEMENTATION

A. Agricultural Data Sources

The platform draws on a mix of structured statistical sources, image datasets, and live environmental APIs rather than a single data provider. ICAR advisories and agricultural research material inform the vector-indexed knowledge base. FAOSTAT and the Directorate of Economics and Statistics (DES) provide agricultural and food statistics used for broader regional context. Bhuvan supplies Indian geospatial reference data. PlantVillage [5] and IP102 [6] are the source image datasets for disease and pest classification respectively. Open-Meteo provides live weather, and NASA POWER provides historical climate data. Sentinel-2 imagery, accessed through the Planetary Computer STAC API, is the satellite data source.

TABLE III.

Data Sources by Function

Source	Function
ICAR	Agricultural advisories, research
FAOSTAT / DES	Agricultural statistics
Bhuvan	Geospatial reference data
PlantVillage [5]	Disease classification images
IP102 [6]	Pest classification images
Open-Meteo	Live weather
NASA POWER	Historical climate
Sentinel-2 / Planetary Computer	Satellite / NDVI

B. Backend and API Design

The backend is implemented in FastAPI, exposing a versioned REST API. Representative endpoints include farm

management, weather, climate, crop recommendation, pest and disease detection, satellite queries, RAG-based question answering, market prices, and mandi routing. A single, consistent response contract — referred to internally as *FarmerResponse*, containing an answer, confidence, an evidence array, warnings, and metadata — is used across the different AI-driven endpoints, so the frontend does not need a separate rendering path for every subsystem's output format.

C. Coordinate Handling

Geospatial coordinate conventions differ between the formats used internally: GeoJSON represents coordinates as [longitude, latitude], while the REST API query parameters use separate named lat and lon fields. This is a common source of integration bugs in geospatial systems generally, and the implementation uses a single shared coordinate-conversion utility specifically to prevent this class of error from being reintroduced independently in different parts of the codebase.

D. Model Routing

Rather than coupling the application to one language-model provider, an AI router layer sits between the safety and evidence layer and the underlying model, capable of directing a request to different providers depending on the task. The reasoning is that different models carry different tradeoffs in reasoning depth, latency, cost, and multilingual quality, and a single fixed choice is unlikely to be optimal across the full range of request types the system handles — a simple factual lookup does not need the same model as a complex multi-hop advisory question, and a voice interaction has a stricter latency budget than a dashboard text query.

VII. EVALUATION METHODOLOGY

The system as described in Sections IV through VI is at the architectural and pre-deployment stage. This section specifies the evaluation protocol intended for use once the individual components and the integrated system are operational, so the evaluation plan is on record independent of eventual results.

A. Machine-Learning Evaluation

The crop-recommendation model is intended to be evaluated using standard classification metrics — accuracy, precision, recall, F1 score, and top-k accuracy, given that a ranked list of plausible crops is arguably more useful than a single top prediction in a domain where several crops may be reasonably suitable for the same conditions.

Pest and disease classification will be evaluated the same way, with two additions specific to the field-versus-lab gap discussed in Section II-B: reporting per-class performance and a confusion matrix rather than aggregate accuracy alone, since aggregate accuracy can mask poor performance on less-represented classes; and reporting accuracy separately on lab-condition benchmark images versus field-captured images, so the gap between the two is measured directly rather than assumed.

B. Retrieval Evaluation

Each retrieval layer's contribution needs to be evaluated both in isolation and as part of the fused system. Standard information-retrieval metrics — precision@k , recall@k , mean

reciprocal rank, and normalized discounted cumulative gain — apply directly to the vector retrieval layer. For the graph layer, evaluation is intended to focus on relationship-query accuracy specifically: whether a graph traversal correctly answers a relationship question a flat retrieval method would either miss or answer incorrectly.

C. Generation Evaluation

Generated responses are intended to be evaluated on faithfulness (does the response match what the retrieved evidence actually supports), answer relevance, context relevance, and citation correctness — whether an evidence citation attached to a claim genuinely supports that claim on manual review. Existing RAG evaluation frameworks such as Ragas are a reasonable starting point for this, adapted to the agricultural domain rather than built from scratch.

D. Hallucination and Abstention Metrics

Two metrics specific to the safety design in Section V are proposed. Unsupported claim rate is the proportion of factual claims in a response not backed by any retrieved evidence:

$$UCR = (\text{unsupported claims}) / (\text{total factual claims}) \times 100$$

Safe abstention accuracy measures whether the system correctly answers questions it has sufficient evidence for and correctly abstains on questions it does not — evaluated against a curated set of queries deliberately including some with insufficient underlying evidence, so the abstention pathway from Section V-C is tested directly rather than only observed incidentally.

TABLE IV.

Planned Evaluation Metrics by Component

Component	Metrics
Crop Recommendation	Accuracy, F1, top-k accuracy
Pest / Disease CV	Per-class F1, confusion matrix, lab-vs-field gap
Vector Retrieval	Precision@k, recall@k, MRR, NDCG
Graph Retrieval	Relationship-query accuracy
Generation	Faithfulness, relevance, citation correctness
Safety Layer	Unsupported claim rate, safe abstention accuracy
System-level	End-to-end latency, availability, retrieval latency

E. Proposed Comparative Experiments

To isolate the contribution of the hybrid retrieval design specifically, four progressively richer configurations are proposed for comparison: an LLM alone with no retrieval; an LLM with vector RAG only; an LLM with vector RAG plus PostgreSQL structured retrieval; and the full proposed system with vector RAG, PostgreSQL, and Neo4j graph retrieval combined. Comparing accuracy, groundedness, evidence coverage, and hallucination rate across these four configurations would show directly whether each additional retrieval layer earns its added complexity, rather than assuming the full hybrid design is better without measuring it against simpler alternatives.

VIII. LIMITATIONS

Several limitations of the current design are worth stating directly.

- Computer-vision models trained substantially on lab-condition image datasets such as PlantVillage are expected to underperform on field-captured images, consistent with the gap documented elsewhere in the literature (Section II-B); this gap has not yet been measured for this system's specific models.
- Crop recommendation and yield-related predictions depend on the quality and regional representativeness of the underlying training data, and models trained on one agricultural region may not transfer cleanly to a different one.
- The system depends on multiple external APIs (Open-Meteo, NASA POWER, Planetary Computer, and others), any of which can experience downtime, rate-limiting, or format changes outside the project's control.
- Satellite-based monitoring is inherently limited by cloud cover, and a field under persistent cloud cover during a critical period may simply have no usable recent imagery.
- The hybrid SQL-graph-vector retrieval design adds real architectural complexity relative to a single-source RAG system, and the evaluation proposed in Section VII-E has not yet been carried out to confirm the added complexity is justified by measurable gains.
- Multilingual support, particularly for Hinglish and regional agricultural terminology, has known difficulty in translation consistency that has not yet been specifically evaluated for this system.
- The evaluation protocol described in Section VII is a plan, not a completed evaluation; no results in this paper should be read as measured outcomes.

IX. FUTURE WORK

Several extensions are planned beyond the current architectural scope. A validated yield-prediction model is intended to connect to the same farm-context and evidence pipeline used elsewhere in the system, rather than being introduced as a disconnected feature. A more dynamic farm digital-twin representation — modeling farm state as it evolves across a season rather than as a static record — is a further planned extension to the farm-centric context model described in Section IV-F.

On the retrieval side, deeper GraphRAG reasoning across crop, soil, weather, disease, pest, region, and advisory relationships simultaneously is planned, extending the two-hop relationship queries the current Neo4j design supports toward genuinely multi-hop reasoning chains. Edge deployment of the vision models is also under consideration, to reduce dependency on connectivity for pest and disease classification in low-connectivity rural areas — a meaningful practical concern given the system's target users.

On the conversational side, expanding Saheta toward a more capable voice-first interface is planned specifically for low-literacy and low-connectivity users, for whom a text-heavy dashboard interface is a real barrier rather than a

minor inconvenience. Federated learning is also being considered as a way to improve model quality using farmer data without centralizing sensitive information, though this remains an early-stage direction rather than a committed design choice. Finally, a genuinely multimodal retrieval system — combining text, images, satellite data, graph relationships, and structured records within a single retrieval mechanism rather than the current design's separate-then-fused approach — represents a longer-term architectural direction beyond what is proposed in Section IV.

X. CONCLUSION

This paper presented the architecture of Prithvi Agro AI, an agricultural intelligence platform built around three ideas that, individually, are not new, but that are rarely combined in one system: hybrid retrieval across structured, relational, and semantic data sources routed per query rather than uniformly; a farm-centric context model that keeps every response scoped to a specific farm's actual live conditions; and a safety and evidence layer that attaches confidence, provenance, and citations to every generated response, with abstention as a legitimate output rather than a failure state to be minimized.

The specific combination of PostgreSQL, Neo4j, and vector retrieval feeding a shared context-fusion step, gated by an explicit evidence-coverage check before language-model reasoning proceeds, is the paper's central architectural contribution. The evaluation methodology in Section VII gives this design a concrete, testable structure — specific metrics, a specific comparative-experiment design isolating what each retrieval layer contributes — rather than leaving the architecture as description alone.

The system is presented here at the design and pre-implementation stage deliberately, without field results attached, since it has not yet been built out and tested at the scale that would make such results meaningful. What this paper offers is the architecture, the reasoning behind combining these particular components rather than others, and an evaluation plan specific enough to hold the eventual implementation accountable to it. The next step is implementation against the specification in Section VI, followed by the evaluation protocol in Section VII, with the extensions in Section IX to follow once that baseline exists.

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